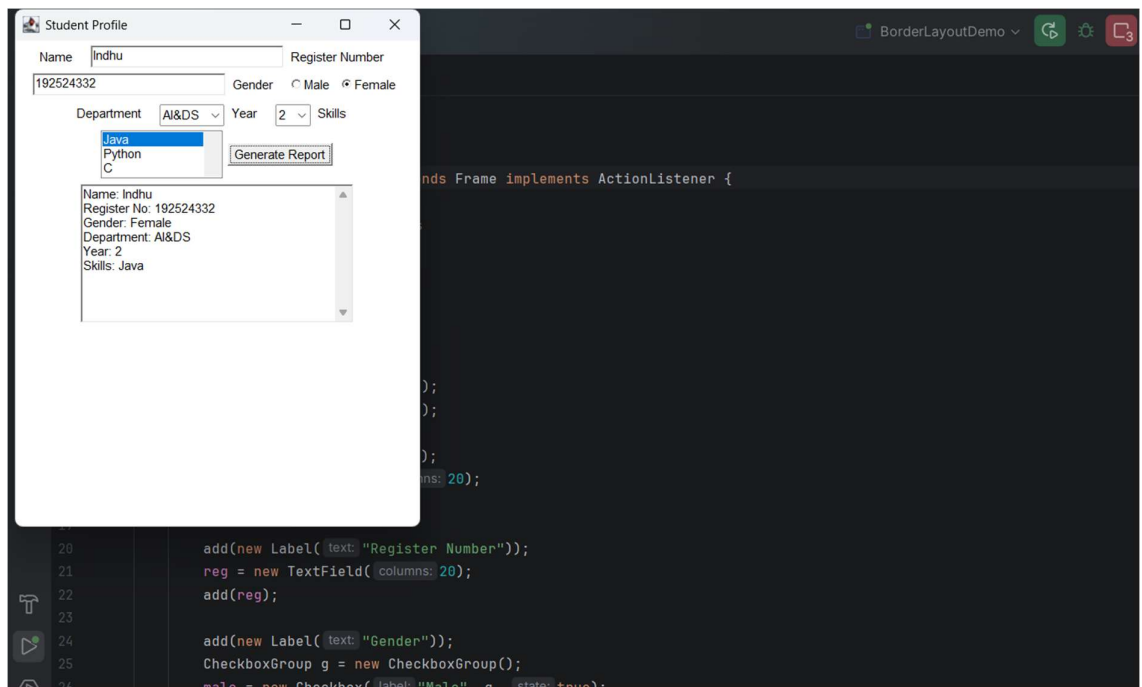


4. Question

```
public class BorderLayoutDemo extends Frame implements ActionListener {  
    TextField name, reg; 3 usages  
    Checkbox male, female; 3 usages  
    Choice dept, year; 7 usages  
    List skills; 7 usages  
    TextArea result; 3 usages  
    Button button; 3 usages  
  
    BorderLayoutDemo() { 1 usage  
        setTitle("Student Profile");  
        setLayout(new FlowLayout());  
  
        add(new Label( text: "Name"));  
        name = new TextField( columns: 20);  
        add(name);  
  
        add(new Label( text: "Register Number"));  
        reg = new TextField( columns: 20);  
        add(reg);  
  
        add(new Label( text: "Gender"));  
        CheckboxGroup g = new CheckboxGroup();  
        male = new Checkbox( label: "Male", g, state: true);  
        female = new Checkbox( label: "Female", g, state: false);  
        add(male);  
        add(female);  
  
        add(new Label( text: "Department"));  
        dept = new Choice();  
        dept.add("CSE");  
        dept.add("AI&DS");
```

```
        add(new Label( text: "Department"));  
        dept = new Choice();  
        dept.add("CSE");  
        dept.add("AI&DS");  
        dept.add("ECE");  
        dept.add("EEE");  
        add(dept);  
  
        add(new Label( text: "Year"));  
        year = new Choice();  
        year.add("1");  
        year.add("2");  
        year.add("3");  
        year.add("4");  
        add(year);  
  
        add(new Label( text: "Skills"));  
        skills = new List( rows: 3, multipleMode: true);  
        skills.add("Java");  
        skills.add("Python");  
        skills.add("C");  
        skills.add("SQL");  
        add(skills);  
  
        button = new Button( label: "Generate Report");  
        button.addActionListener( l: this);  
        add(button);
```

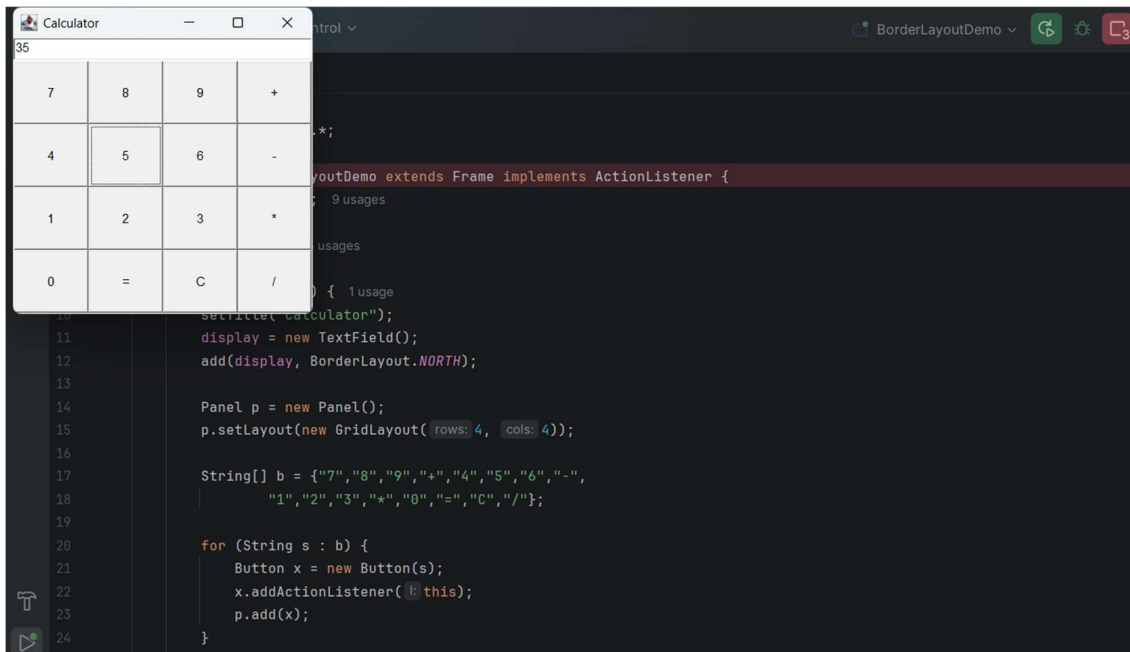
```
BorderLayoutDemo.java x Run / Debug Configur...
4 public class BorderLayoutDemo extends Frame implements ActionListener {
12     BorderLayoutDemo() { // 1 usage
13         skills.add("Java");
53         add(skills);
54
55         button = new Button( label: "Generate Report");
56         button.addActionListener( !: this);
57         add(button);
58
59         result = new TextArea( rows: 8, columns: 30);
60         add(result);
61
62         setSize( width: 400, height: 500);
63         setVisible(true);
64     }
65
66     public void actionPerformed(ActionEvent e) {
67         result.setText("Name: " + name.getText() +
68             "\nRegister No: " + reg.getText() +
69             "\nGender: " + (male.getState() ? "Male" : "Female") +
70             "\nDepartment: " + dept.getSelectedItem() +
71             "\nYear: " + year.getSelectedItem() +
72             "\nSkills: " + skills.getSelectedItems()[0]);
73     }
74
75     public static void main(String[] args) {
76         new BorderLayoutDemo();
77     }
78 }
```



5. Question

```
BorderLayoutDemo.java x
1 import java.awt.*;
2 import java.awt.event.*;
3
4 public class BorderLayoutDemo extends Frame implements ActionListener {
5     TextField display; 9 usages
6     double a; 5 usages
7     String op = ""; 5 usages
8
9     BorderLayoutDemo() { 1 usage
10         setTitle("Calculator");
11         display = new TextField();
12         add(display, BorderLayout.NORTH);
13
14         Panel p = new Panel();
15         p.setLayout(new GridLayout( rows: 4, cols: 4));
16
17         String[] b = {"7","8","9","+","4","5","6","-",
18                     "1","2","3","*","0","=","C","/"};
19
20         for (String s : b) {
21             Button x = new Button(s);
22             x.addActionListener( this);
23             p.add(x);
24         }
25
26         add(p);
27         setSize( width: 300, height: 300);
28         setVisible(true);
29     }
30
31     public void actionPerformed(ActionEvent e) {
```

```
BorderLayoutDemo.java x
4 public class BorderLayoutDemo extends Frame implements ActionListener {
7     BorderLayoutDemo() { 1 usage
29     }
30
31     public void actionPerformed(ActionEvent e) {
32         String s = e.getActionCommand();
33
34         if (s.matches( regex: "[0-9]")) {
35             display.setText(display.getText() + s);
36         } else if ("+-*/".contains(s)) {
37             a = Double.parseDouble(display.getText());
38             op = s;
39             display.setText("");
40         } else if (s.equals("=")) {
41             double b = Double.parseDouble(display.getText());
42             double r = 0;
43
44             if (op.equals("+")) r = a + b;
45             if (op.equals("-")) r = a - b;
46             if (op.equals("*")) r = a * b;
47             if (op.equals("/")) r = a / b;
48
49             display.setText("" + r);
50         } else {
51             display.setText("");
52         }
53     }
54
55     public static void main(String[] args) {
56         new BorderLayoutDemo();
57     }
58 }
```

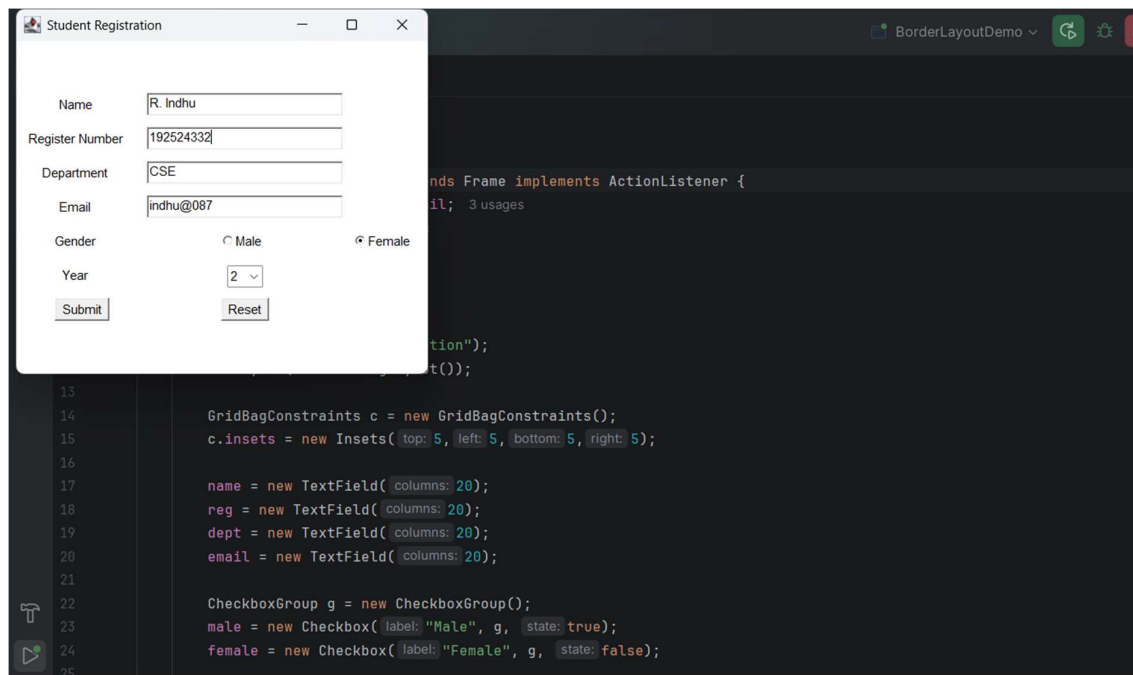


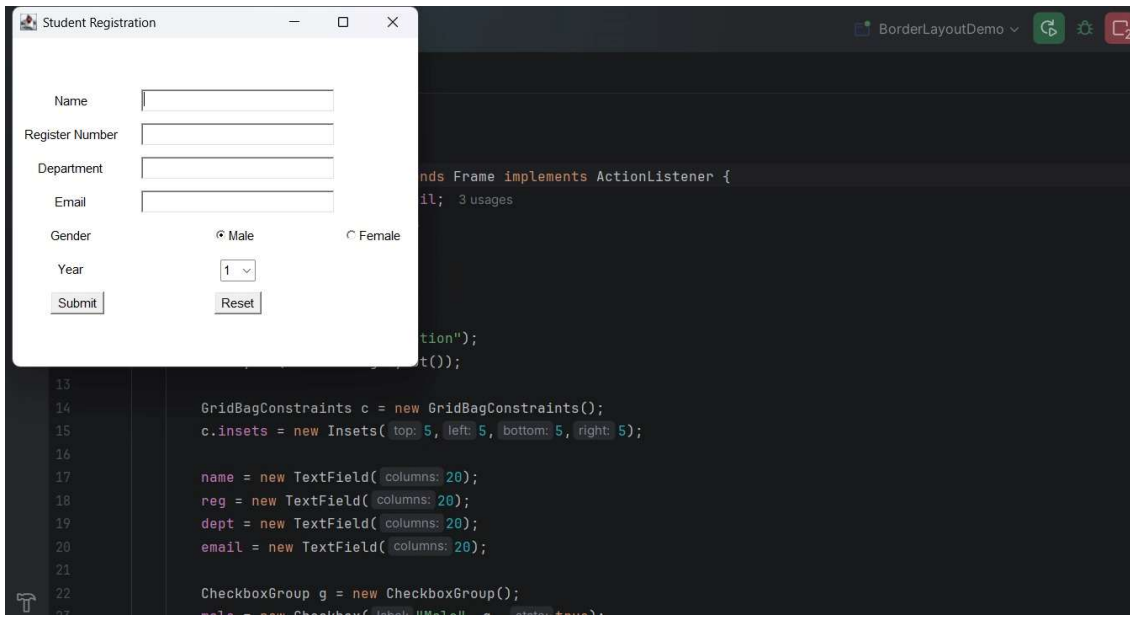
6. Question

```
BorderLayoutDemo.java x
1 import java.awt.*;
2 import java.awt.event.*;
3
4 public class BorderLayoutDemo extends Frame implements ActionListener {
5     TextField name, reg, dept, email; 3 usages
6     Checkbox male, female; 2 usages
7     Choice year; 6 usages
8     Button submit, reset; 4 usages
9
10    BorderLayoutDemo() { 1 usage
11        setTitle("Student Registration");
12        setLayout(new GridBagLayout());
13
14        GridBagConstraints c = new GridBagConstraints();
15        c.insets = new Insets( top: 5, left: 5, bottom: 5, right: 5);
16
17        name = new TextField( columns: 20);
18        reg = new TextField( columns: 20);
19        dept = new TextField( columns: 20);
20        email = new TextField( columns: 20);
21
22        CheckboxGroup g = new CheckboxGroup();
23        male = new Checkbox( label: "Male", g, state: true);
24        female = new Checkbox( label: "Female", g, state: false);
25
26        year = new Choice();
27        year.add("1");
28        year.add("2");
29        year.add("3");
30        year.add("4");
31    }
```

```
LL linked list v Version Control v BorderLayoutDemo v
BorderLayoutDemo.java x
4 public class BorderLayoutDemo extends Frame implements ActionListener {
10    BorderLayoutDemo() { 1 usage
33        addItem(name, c, a: 1, b: 0);
34
35        addItem(new Label( text: "Register Number"), c, a: 0, b: 1);
36        addItem(reg, c, a: 1, b: 1);
37
38        addItem(new Label( text: "Department"), c, a: 0, b: 2);
39        addItem(dept, c, a: 1, b: 2);
40
41        addItem(new Label( text: "Email"), c, a: 0, b: 3);
42        addItem(email, c, a: 1, b: 3);
43
44        addItem(new Label( text: "Gender"), c, a: 0, b: 4);
45        addItem(male, c, a: 1, b: 4);
46        addItem(female, c, a: 2, b: 4);
47
48        addItem(new Label( text: "Year"), c, a: 0, b: 5);
49        addItem(year, c, a: 1, b: 5);
50
51        submit = new Button( label: "Submit");
52        reset = new Button( label: "Reset");
53
54        submit.addActionListener( l: this);
55        reset.addActionListener( l: this);
56
57        addItem(submit, c, a: 0, b: 6);
58        addItem(reset, c, a: 1, b: 6);
59
60        setSize( width: 400, height: 350);
61        setVisible(true);
62    }
```

```
BorderLayoutDemo.java x
4 public class BorderLayoutDemo extends Frame implements ActionListener {
5     BorderLayoutDemo() { 1 usage
6         setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
7     }
8
9     @
10    void addItem(Component x, GridBagConstraints c, int a, int b) { 15 usages
11        c.gridx = a;
12        c.gridy = b;
13        add(x, c);
14    }
15
16    @
17    public void actionPerformed(ActionEvent e) {
18        if (e.getSource() == submit)
19            System.out.println("Registration Submitted");
20        else {
21            name.setText("");
22            reg.setText("");
23            dept.setText("");
24            email.setText("");
25        }
26    }
27
28    public static void main(String[] args) {
29        new BorderLayoutDemo();
30    }
31
32 }
```

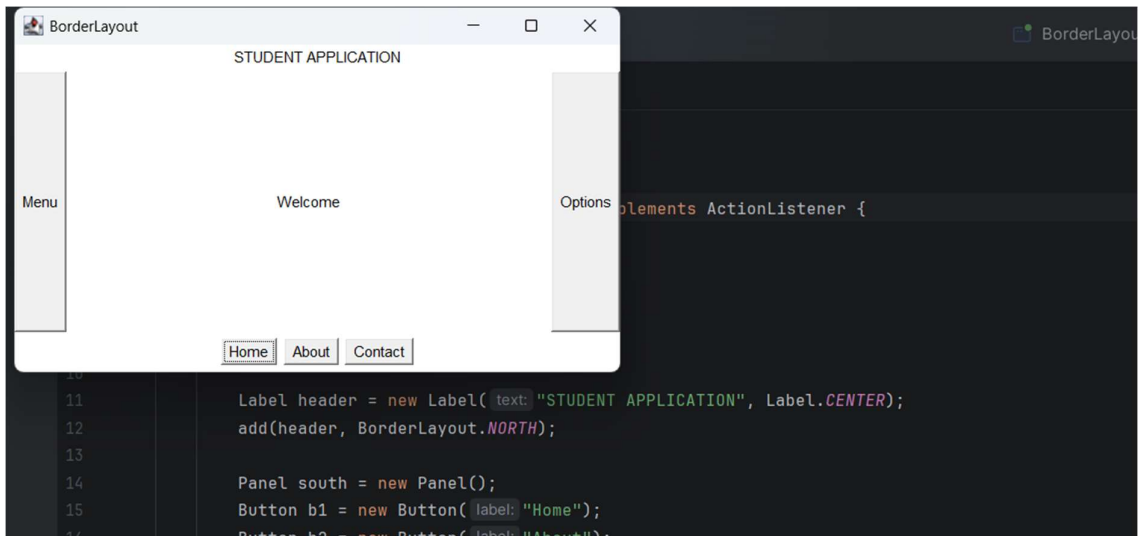




7. Question

```
BorderLayoutDemo.java x
1  import java.awt.*;
2  import java.awt.event.*;
3
4  public class BorderLayoutDemo extends Frame implements ActionListener {
5      Label center; 3 usages
6
7      BorderLayoutDemo() { 1 usage
8          setTitle("BorderLayout");
9          setLayout(new BorderLayout());
10
11          Label header = new Label(text: "STUDENT APPLICATION", Label.CENTER);
12          add(header, BorderLayout.NORTH);
13
14          Panel south = new Panel();
15          Button b1 = new Button(label: "Home");
16          Button b2 = new Button(label: "About");
17          Button b3 = new Button(label: "Contact");
18
19          b1.addActionListener(l: this);
20          b2.addActionListener(l: this);
21          b3.addActionListener(l: this);
22
23          south.add(b1);
24          south.add(b2);
25          south.add(b3);
26          add(south, BorderLayout.SOUTH);
27
28          add(new Button(label: "Options"), BorderLayout.EAST);
29          add(new Button(label: "Menu"), BorderLayout.WEST);
30
31
32
33
34
35
36
37
38  public void actionPerformed(ActionEvent e) {
39      center.setText(e.getActionCommand() + " Selected");
40  }
41
42  public static void main(String[] args) {
43      new BorderLayoutDemo();
44  }
45  }
46
```

```
BorderLayoutDemo.java x
4  public class BorderLayoutDemo extends Frame implements ActionListener {
7      BorderLayoutDemo() { 1 usage
23          south.add(b1);
24          south.add(b2);
25          south.add(b3);
26          add(south, BorderLayout.SOUTH);
27
28          add(new Button(label: "Options"), BorderLayout.EAST);
29          add(new Button(label: "Menu"), BorderLayout.WEST);
30
31          center = new Label(text: "Welcome", Label.CENTER);
32          add(center, BorderLayout.CENTER);
33
34          setSize(width: 500, height: 300);
35          setVisible(true);
36      }
37
38  public void actionPerformed(ActionEvent e) {
39      center.setText(e.getActionCommand() + " Selected");
40  }
41
42  public static void main(String[] args) {
43      new BorderLayoutDemo();
44  }
45  }
46
```



8. Question

```
import java.awt.*;
import java.awt.event.*;

class CardLayoutDemo extends Frame implements ActionListener {
    CardLayout card; 4 usages
    Panel cards; 7 usages

    CardLayoutDemo() { 1 usage
        setTitle("Student Information");
        card = new CardLayout();
        cards = new Panel(card);

        Panel p1 = new Panel();
        p1.add(new Label(text: "Card 1: Student Information"));
        p1.add(new TextField(text: "Name", columns: 15));

        Panel p2 = new Panel();
        p2.add(new Label(text: "Card 2: Academic Information"));
        p2.add(new TextField(text: "Department", columns: 15));

        Panel p3 = new Panel();
        p3.add(new Label(text: "Card 3: Confirmation / Report"));

        cards.add(p1, constraints: "1");
        cards.add(p2, constraints: "2");
        cards.add(p3, constraints: "3");

        add(cards, BorderLayout.CENTER);

        Panel buttons = new Panel();
```

```
java.java x
4      class CardLayoutDemo extends Frame implements ActionListener {
8          CardLayoutDemo() { 1 usage
33              Button next = new Button(label: "Next");
34              Button submit = new Button(label: "Submit");
35
36              prev.addActionListener(this);
37              next.addActionListener(this);
38              submit.addActionListener(this);
39
40              buttons.add(prev);
41              buttons.add(next);
42              buttons.add(submit);
43
44              add(buttons, BorderLayout.SOUTH);
45
46              setSize(width: 500, height: 300);
47              setVisible(true);
48          }
49
50          public void actionPerformed(ActionEvent e) {
51              if (e.getActionCommand().equals("Next"))
52                  card.next(cards);
53              else if (e.getActionCommand().equals("Previous"))
54                  card.previous(cards);
55              else
56                  System.out.println("Submitted");
57          }
58
59          public static void main(String[] args) {
60              new CardLayoutDemo();
61          }
```

